Name: Muhammad Luqman
Roll No: FA24-BCS-074-B

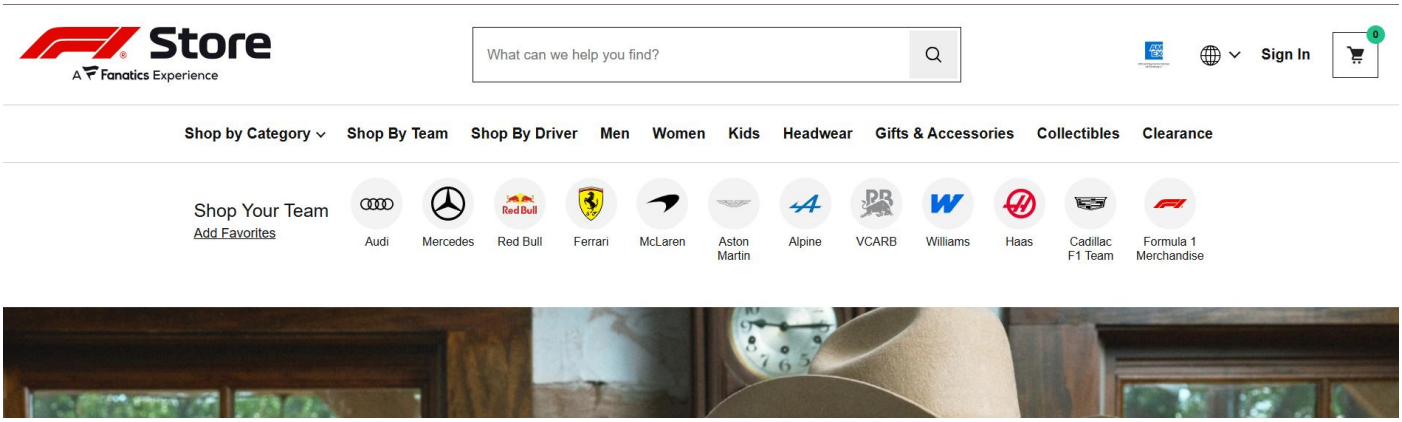
Statement:

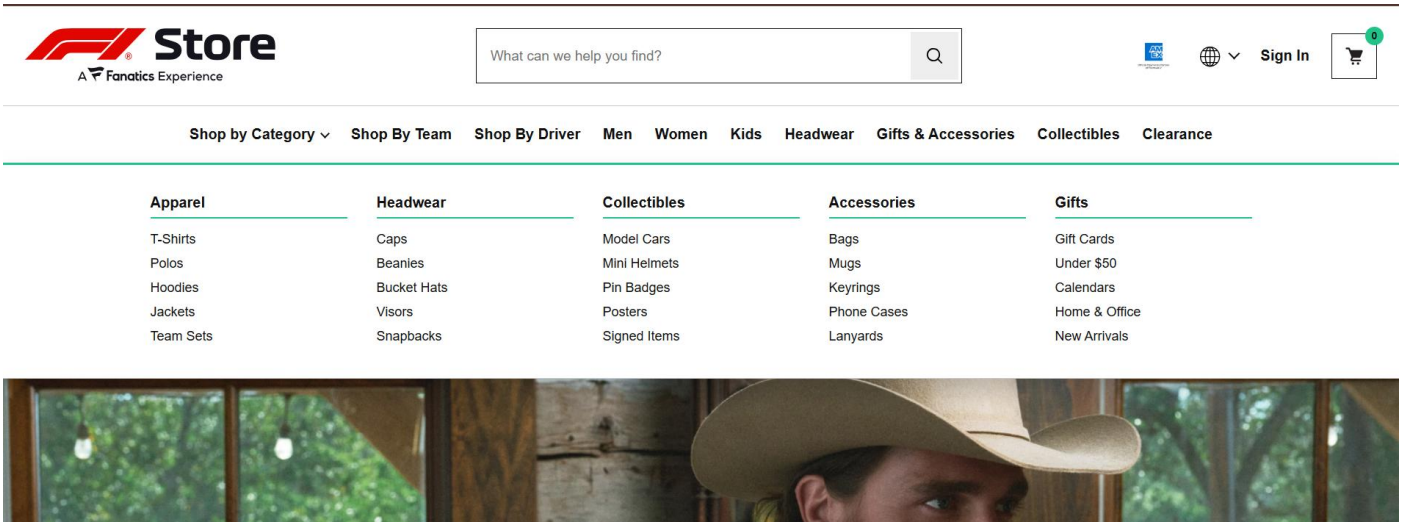
Lab Mid Term - Upgrade Your E-commerce Navigation, Checkout and Cart

Difficulty: Medium · Covers: Lab 1-3, Assignment 1-2 (Lectures 3-13: semantic HTML, forms, CSS box model, positioning, CSS3, Flexbox and Grid, responsive design and frameworks, core JavaScript, array methods, DOM and events) · Folder: midterm/

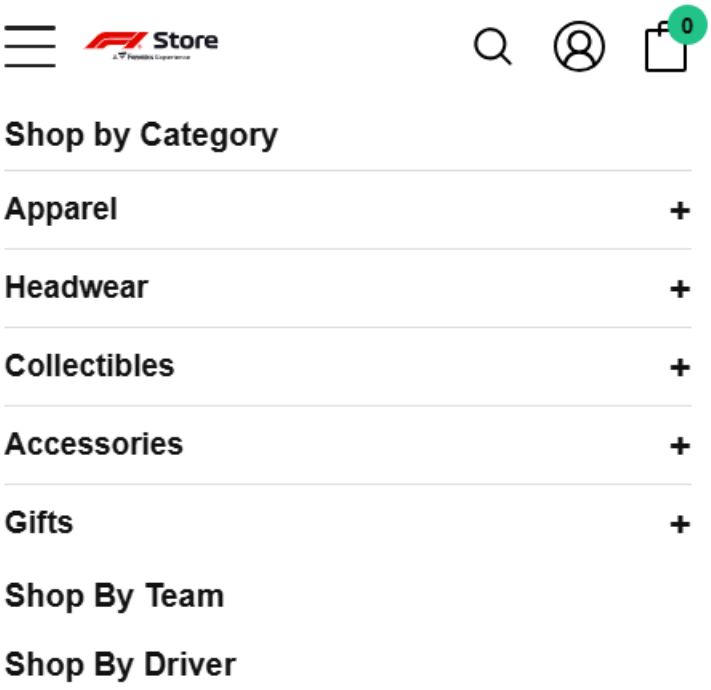
This is an open-book, open-internet, AI-allowed exam. You may use ChatGPT, Claude, Copilot, Cursor or any other AI agent. What is being tested is whether you can take your own term project and change it correctly under time pressure, and whether you understand what was changed. Every task below depends on your own existing markup and on numbers derived from your roll number, so a generic answer copied from an AI or a classmate will not satisfy it.

Images:

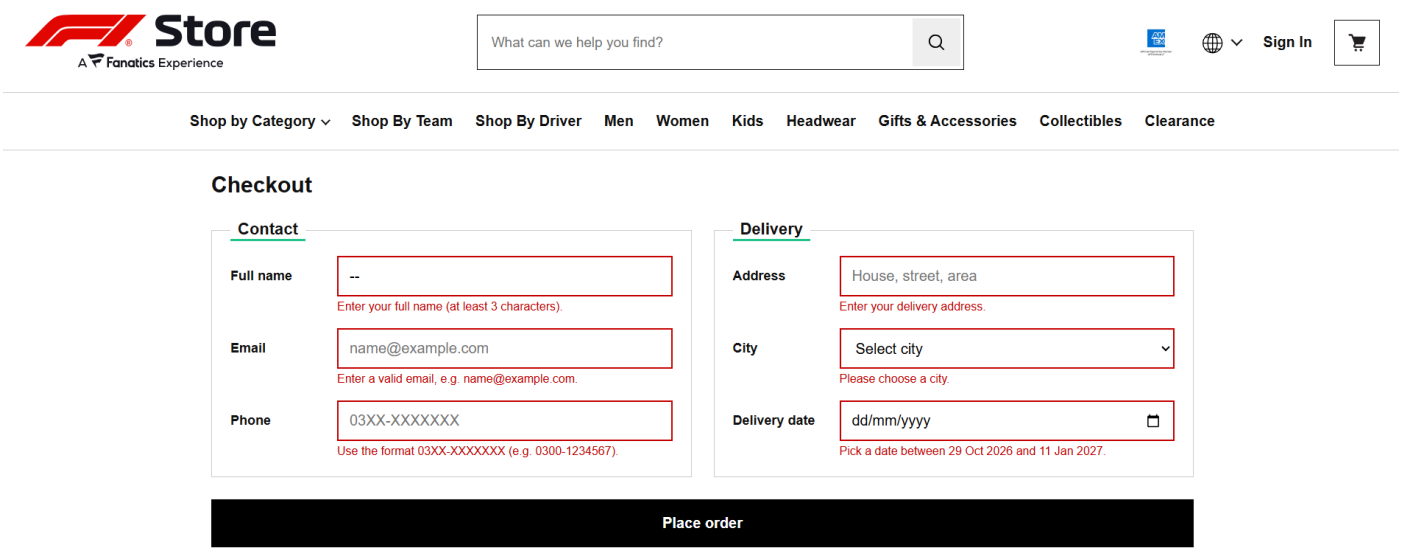




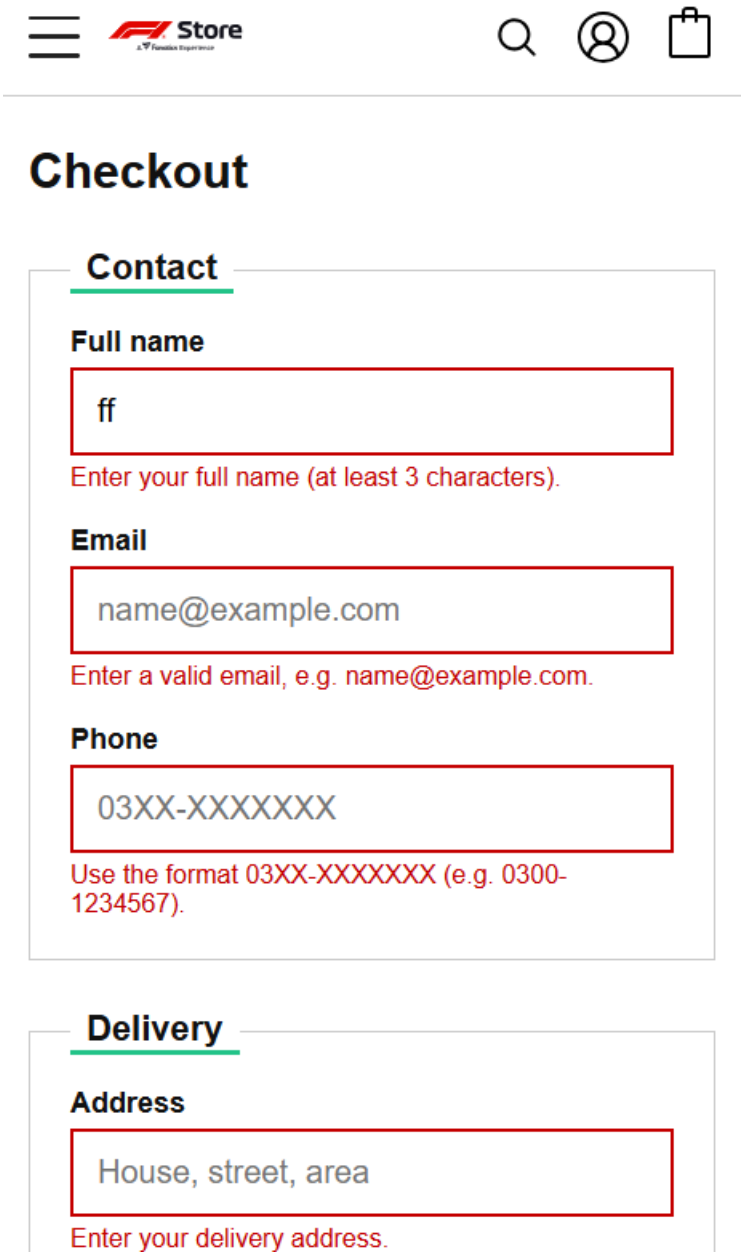
Task 1-Shop by category 1200 bp



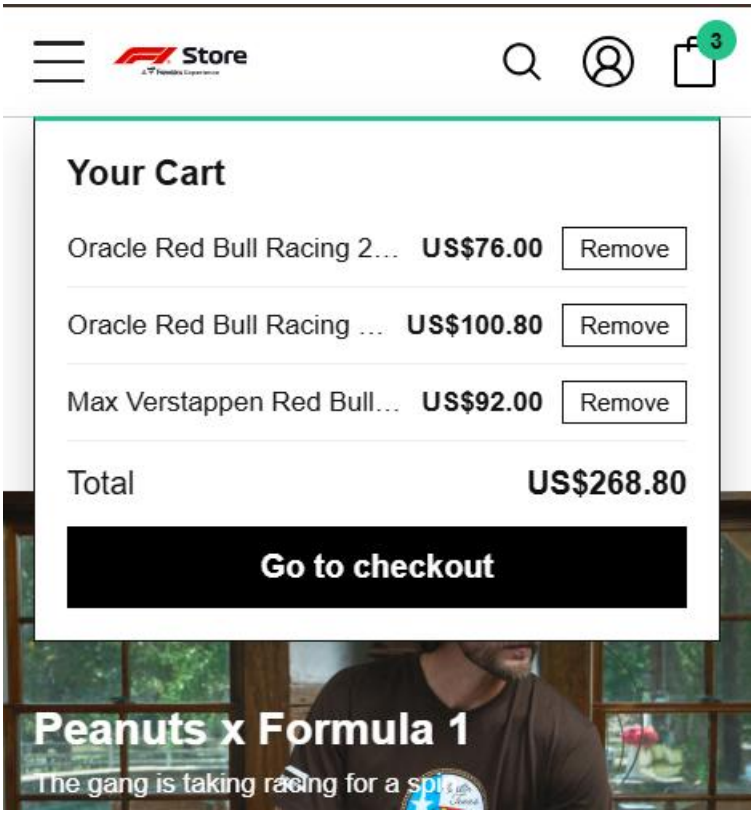
Task 1-Shop by category 375 bp



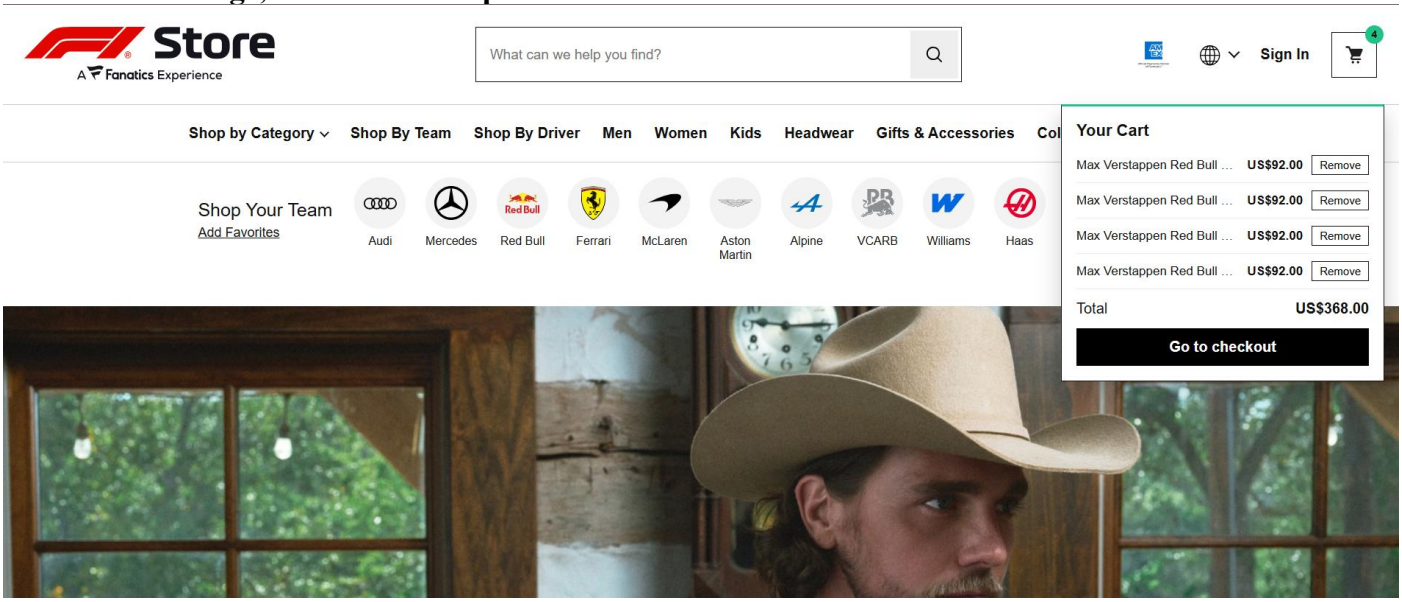
Task 2-Checkout Validation 1200bp



Task 2-Checkout Validation 375bp



Task 3-Cart badge, mini cart 375 bp



Task 3-Cart badge, mini cart 1200 bp

Different Code:
Task 1 - Mega-menu in the navbar
New file: mega-menu.css (full file)

```
:root {
  --accent: hsl(158, 70%, 45%); /* hsl(H, 70%, 45%) with H = 158 */
}

.bp-up {
  display: none !important;
}
@media (min-width: 674px) {
```

```
  .bp-up {
    display: flex !important;
  }
  .bp-down {
    display: none !important;
  }
}

/* ---- Stacking: header (and the panel inside it) above all page content -- */
.header {
  position: relative;
  z-index: 1000;
}
.category-nav {
  position: relative; /* the mega panel is positioned against the navbar */
  z-index: 1000;
}

/* ---- Navbar expands at BP instead of Bootstrap's 768px ---------- */
@media (min-width: 674px) {
  .category-nav .navbar-collapse.collapse {
    display: flex !important; /* beats Bootstrap's .collapse:not(.show) */
    flex-basis: auto;
  }
  .category-nav .navbar-nav {
    flex-direction: row;
  }
  .hamburger {
    display: none;
  }
  .header-main {
    flex-wrap: nowrap;
  }
}

@media (min-width: 674px) and (max-width: 768px) {
  .header-main {
    padding: 1.5rem;
    justify-content: space-between;
  }
  .f1-logo {
    flex: 1;
    height: 3.75rem;
    width: auto;
  }
  .search-bar {
    width: 45%;
    margin-top: 0;
  }
  .category-nav {
    flex-wrap: wrap;
    justify-content: center;
    gap: 0.5rem 1.5rem;
    padding: 0.65rem 1rem;
    border-top: 1px solid #ccc;
```

```
  }
  .category-nav .nav-link {
    padding: 0;
  }
}

.mega-trigger::after {
  content: "";
  display: none;
  width: 0.45rem;
  height: 0.45rem;
  margin-left: 0.45rem;
  border-right: 2px solid #111;
  border-bottom: 2px solid #111;
  transform: translateY(-0.15rem) rotate(45deg);
}

.mega-panel {
  display: none; /* below BP the accordion is used instead */
}

@media (min-width: 674px) {
  .mega-trigger::after {
    display: inline-block;
  }

  .mega-panel {
    display: block;
    position: absolute; /* out of flow: does not push the page down */
    top: 100%; /* directly under the navbar */
    left: 0;
    right: 0; /* full width */
    z-index: 1000;
    background: #fff;
    border-top: 3px solid var(--accent);
    border-bottom: 1px solid #ccc;
    box-shadow: 0 14px 24px rgba(0, 0, 0, 0.18);
    padding: 1.5rem 1rem;

    /* closed state: not focusable, not hoverable, invisible */
    visibility: hidden;
    opacity: 0;
    transition:
      opacity 0.15s ease,
      visibility 0s linear 0.15s;
  }

  .mega-item:hover > .mega-panel,
  .mega-item:focus-within > .mega-panel {
    visibility: visible;
    opacity: 1;
    transition-delay: 0s;
  }

  .mega-panel::before {
```

```
    content: "";
    position: absolute;
    left: 0;
    right: 0;
    top: -0.65rem;
    height: 0.65rem;
  }

  .mega-grid {
    display: grid;
    grid-template-columns: repeat(5, 1fr); /* K = 5 columns */
    gap: 1rem;
    max-width: 1200px;
    margin: 0 auto;
  }

  .mega-heading {
    margin: 0 0 0.6rem;
    padding-bottom: 0.4rem;
    font-size: 1rem;
    font-weight: 700;
    border-bottom: 2px solid var(--accent);
  }

  .mega-col ul {
    list-style: none;
    margin: 0;
    padding: 0;
  }

  .category-nav .mega-col a {
    display: block;
    padding: 0.3rem 0;
    font-size: 0.9rem;
    font-weight: 400;
    white-space: normal;
  }
  .category-nav .mega-col a:hover,
  .category-nav .mega-col a:focus-visible {
    text-decoration: underline;
    text-decoration-color: var(--accent);
    text-decoration-thickness: 2px;
    text-underline-offset: 3px;
  }

  .mega-accordion {
    display: none;
  }
}

@media (min-width: 992px) {
  .category-nav {
    padding-left: 2rem;
    padding-right: 2rem;
  }
```

```
  .category-nav .navbar-nav {
    column-gap: clamp(0.9rem, 1.6vw, 2.5rem);
  }

  .mega-panel::before {
    top: -1.35rem;
    height: 1.35rem;
  }
}

@media (min-width: 1000px) {
  .mega-grid {
    gap: 2rem;
  }
  .mega-panel {
    padding: 2rem;
  }
}

.category-nav a:focus-visible,
.mega-accordion summary:focus-visible {
  outline: 2px solid var(--accent);
  outline-offset: 2px;
}

.mega-item {
  width: 100%;
}
@media (min-width: 674px) {
  .mega-item {
    width: auto;
  }
}

.mega-accordion details {
  border-top: 1px solid #ddd;
}
.mega-accordion summary {
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 0.6rem 0;
  list-style: none;
  cursor: pointer;
  font-size: 0.95rem;
  font-weight: 600;
}
.mega-accordion summary::-webkit-details-marker {
  display: none;
}
.mega-accordion summary::after {
  content: "+";
  font-size: 1.2rem;
  line-height: 1;
}
```

```
.mega-accordion details[open] summary::after {
  content: "\2013"; /* en dash */
}
.mega-accordion details[open] > summary {
  border-left: 3px solid var(--accent);
  padding-left: 0.6rem;
}
.mega-accordion ul {
  list-style: none;
  margin: 0;
  padding: 0 0 0.5rem 1rem;
}
.category-nav .mega-accordion a {
  display: block;
  padding: 0.35rem 0;
  font-size: 0.9rem;
  font-weight: 400;
  white-space: normal;
}
```

index.html - changes for Task 1

Stylesheet link in <head> (the cart.css link next to it belongs to Task 3):

```
+    <link rel="stylesheet" href="mega-menu.css" />
```

Bootstrap d-md-* helpers replaced by the BP-based bp-up / bp-down classes (switch at 674px instead of 768px):

```
-        <div class="search-bar d-none d-md-flex">
+        <div class="search-bar bp-up">
-        <div class="header-icons d-md-none">
+        <div class="header-icons bp-down">
-        <div class="header-right d-none d-md-flex">
+        <div class="header-right bp-up">
```

<nav> no longer uses navbar-expand-md (the expansion at BP is done in mega-menu.css), and gets an accessible name:

```
-      <nav class="category-nav navbar navbar-expand-md">
+      <nav class="category-nav navbar" aria-label="Main navigation">
```

Mobile-only links switched from d-md-none to bp-down:

```
-        <li class="nav-item d-md-none"><a class="nav-link" href="#">Gift Cards</a></li>
-        <li class="nav-item d-md-none"><a class="nav-link" href="#">Track Order</a></li>
-        <li class="nav-item d-md-none"><a class="nav-link" href="#">Help</a></li>
-        <li class="nav-item d-md-none"><a class="nav-link" href="#">My Account</a></li>
+        <li class="nav-item bp-down"><a class="nav-link" href="#">Gift Cards</a></li>
+        <li class="nav-item bp-down"><a class="nav-link" href="#">Track Order</a></li>
+        <li class="nav-item bp-down"><a class="nav-link" href="#">Help</a></li>
+        <li class="nav-item bp-down"><a class="nav-link" href="#">My Account</a></li>
```

New "Shop by Category" (added as the first item of the <ul class="navbar-nav">, lines 57-169). It contains the 5-column panel (a <ul> per column with an <h3> heading) for BP and wider, and the <details>/<summary> accordion for below BP:

```
          <li class="nav-item mega-item">
            <a class="nav-link mega-trigger" href="#" aria-haspopup="true">Shop by Category</a>

            <!-- BP and wider: full-width panel, 5 columns (hover / :focus-within) -->
            <div class="mega-panel" id="mega-panel">
              <div class="mega-grid">
                <div class="mega-col">
                  <h3 class="mega-heading">Apparel</h3>
                  <ul>
```

```
        <li><a href="#">T-Shirts</a></li>
        <li><a href="#">Polos</a></li>
        <li><a href="#">Hoodies</a></li>
        <li><a href="#">Jackets</a></li>
        <li><a href="#">Team Sets</a></li>
      </ul>
    </div>
    <div class="mega-col">
      <h3 class="mega-heading">Headwear</h3>
      <ul>
        <li><a href="#">Caps</a></li>
        <li><a href="#">Beanies</a></li>
        <li><a href="#">Bucket Hats</a></li>
        <li><a href="#">Visors</a></li>
        <li><a href="#">Snapbacks</a></li>
      </ul>
    </div>
    <div class="mega-col">
      <h3 class="mega-heading">Collectibles</h3>
      <ul>
        <li><a href="#">Model Cars</a></li>
        <li><a href="#">Mini Helmets</a></li>
        <li><a href="#">Pin Badges</a></li>
        <li><a href="#">Posters</a></li>
        <li><a href="#">Signed Items</a></li>
      </ul>
    </div>
    <div class="mega-col">
      <h3 class="mega-heading">Accessories</h3>
      <ul>
        <li><a href="#">Bags</a></li>
        <li><a href="#">Mugs</a></li>
        <li><a href="#">Keyrings</a></li>
        <li><a href="#">Phone Cases</a></li>
        <li><a href="#">Lanyards</a></li>
      </ul>
    </div>
    <div class="mega-col">
      <h3 class="mega-heading">Gifts</h3>
      <ul>
        <li><a href="#">Gift Cards</a></li>
        <li><a href="#">Under $50</a></li>
        <li><a href="#">Calendars</a></li>
        <li><a href="#">Home &amp;amp; Office</a></li>
        <li><a href="#">New Arrivals</a></li>
      </ul>
    </div>
  </div>
</div>

<!-- below BP: same 5 groups as an accordion inside the hamburger menu -->
<div class="mega-accordion">
  <details>
    <summary>Apparel</summary>
    <ul>
```

```
        <li><a href="#">T-Shirts</a></li>
        <li><a href="#">Polos</a></li>
        <li><a href="#">Hoodies</a></li>
        <li><a href="#">Jackets</a></li>
        <li><a href="#">Team Sets</a></li>
      </ul>
    </details>
    <details>
      <summary>Headwear</summary>
      <ul>
        <li><a href="#">Caps</a></li>
        <li><a href="#">Beanies</a></li>
        <li><a href="#">Bucket Hats</a></li>
        <li><a href="#">Visors</a></li>
        <li><a href="#">Snapbacks</a></li>
      </ul>
    </details>
    <details>
      <summary>Collectibles</summary>
      <ul>
        <li><a href="#">Model Cars</a></li>
        <li><a href="#">Mini Helmets</a></li>
        <li><a href="#">Pin Badges</a></li>
        <li><a href="#">Posters</a></li>
        <li><a href="#">Signed Items</a></li>
      </ul>
    </details>
    <details>
      <summary>Accessories</summary>
      <ul>
        <li><a href="#">Bags</a></li>
        <li><a href="#">Mugs</a></li>
        <li><a href="#">Keyrings</a></li>
        <li><a href="#">Phone Cases</a></li>
        <li><a href="#">Lanyards</a></li>
      </ul>
    </details>
    <details>
      <summary>Gifts</summary>
      <ul>
        <li><a href="#">Gift Cards</a></li>
        <li><a href="#">Under $50</a></li>
        <li><a href="#">Calendars</a></li>
        <li><a href="#">Home &amp;amp; Office</a></li>
        <li><a href="#">New Arrivals</a></li>
      </ul>
    </details>
  </div>
</li>
```